

Name	CSU ID #
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Be sure to read each question carefully. You must choose and answer **exactly two** of the four problems. If you attempt more than two, only the first two will be graded. Write your final answers in the boxes provided. Only answers written in the boxes will be graded. You may use the surrounding space for scratch work. Each problem is worth the same number of points.

1. The state variables of an ideal gas satisfy

$$PV = nRT,$$

where n and R are constants. Compute

$$\left(\frac{\partial V}{\partial T}\right) \left(\frac{\partial T}{\partial P}\right) \left(\frac{\partial P}{\partial V}\right).$$

2. Let $z = f(u, v)$, where f is twice differentiable, and let

$$u = 3x^2y, \quad v = 4xy^2.$$

Compute $\frac{\partial^2 z}{\partial x^2}$ in terms of x , y , and the partial derivatives f_u , f_{uu} , f_{uv} , and f_{vv} .

3. Let $z = f(x, y)$, where

$$x = t^2 + 1, \quad y = 2t - 1.$$

Given that $\nabla f(2, 1) = (3, -2)$, compute $\frac{dz}{dt}$ at $t = 1$.

4. The temperature in a room is

$$T(x, y, z) = 9x^2 - 3y^2 + 6xyz.$$

A fly is at $(2, 2, 2)$. Find the unit vector in the direction the fly should move so that the temperature decreases most rapidly.